



ARKA JAIN
University
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ONLINE EXAM SYSTEM

PROJECT PROGRESS SEMINAR 3

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CONTENT

- 1.) About The Project.
- 2.) System Requirement.
- 3.) Data flow diagram.
- 3.) Components..
- 4.) Features
- 5.) Screenshot of the Project.
- 6.) Conclusion



About The Project

The Online Examination Application is a comprehensive web-based platform designed to facilitate the seamless administration, delivery, and management of online exams.

This application which has been made using MERN stack (MongoDB, Express.js, React and Node.js) aims at providing an educational institutions, corporate training programs and certification authorities with an efficient, scalable and user-friendly Experience.

The scope of an online exam system project includes user management (admins, teachers, students), exam creation and scheduling, question bank management, secure exam conduction with proctoring, automated grading, and detailed result reporting. It should support multiple question types, randomization, and accessibility across devices. Security measures, including data protection and cheating prevention, are crucial.



HARDWARE REQUIREMENT

► Client-Side (For Students & Examiners)

- **Processor:** Intel Core i5 (minimum) or AMD equivalent
- **RAM:** 4GB (minimum), 8GB (recommended)
- **Storage:** 20GB free space (for exam-related files and caching)
- **Display:** Minimum 1280x720 resolution
- **Internet Connection:** Minimum 5 Mbps for stable connectivity
- **Peripherals:**
 - Webcam (for proctored exams)
 - Microphone (if voice recording is required)
 - Keyboard & Mouse (if using a desktop)

SOFTWARE REQUIREMENT

1. Operating System

- **For Server** : Windows Server 2016 or later
- **For Client Devices (Students & Examiners):**
- Windows 10/11, macOS, or Linux

2. Web Server

- Apache HTTP Server
- Nginx (for high-performance systems)
- IIS (for Windows-based deployments)

3. Database Management System (DBMS)

MongoDB

4. Programming Languages & Frameworks

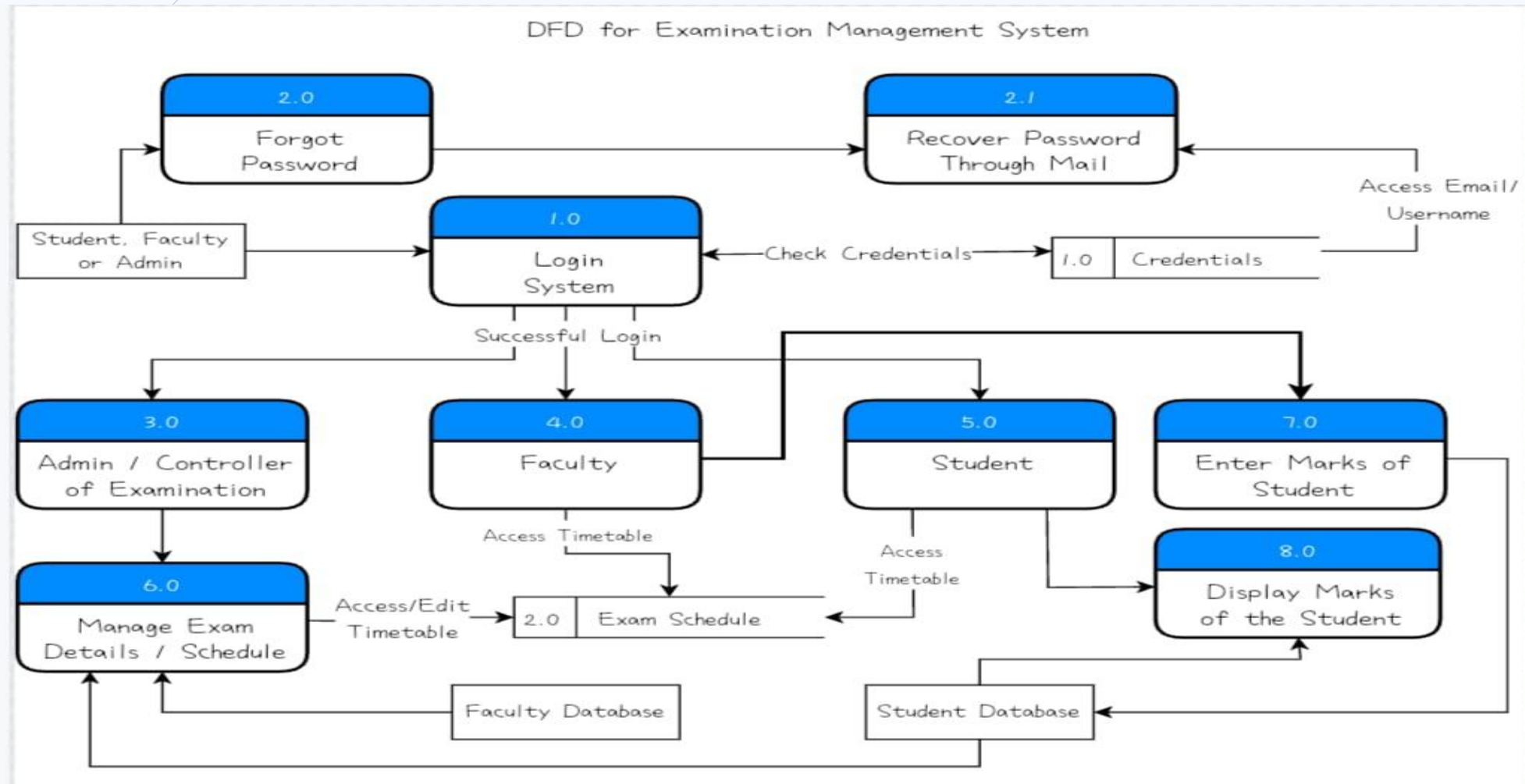
•Backend:

Node.js with Express.js

Frontend:

- HTML, CSS, JavaScript

Data flow diagram



KEY COMPONENTS (USER/ADMIN)

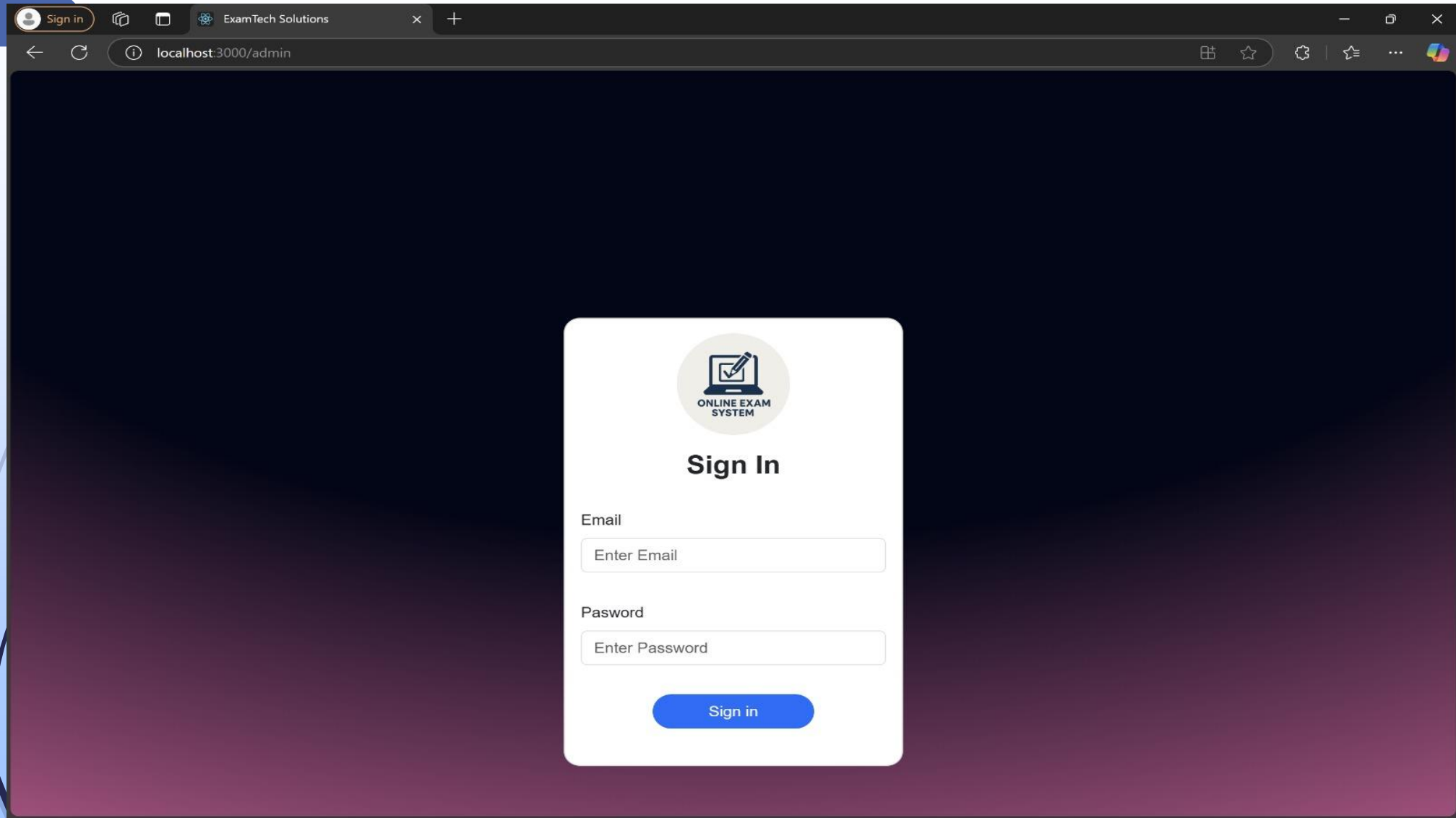
USER

- - Take test
- - View result
- - Login
- Automatic logout

ADMIN

-
- - Manage subjects, exams, questions (Add/Delete/Update)
- - View all/individual results
- - View exam questions
- - Login/Logout

HOME PAGE



The screenshot shows a web browser window with the address bar displaying 'localhost:3000/admin'. The page features a dark blue gradient background. In the center, there is a white rounded rectangle containing the 'Sign In' form. At the top of this rectangle is a circular logo with a laptop icon and the text 'ONLINE EXAM SYSTEM'. Below the logo, the text 'Sign In' is displayed in a bold font. The form includes two input fields: 'Email' with a placeholder 'Enter Email' and 'Pasword' (misspelled) with a placeholder 'Enter Password'. A blue 'Sign in' button is located at the bottom of the form.

Sign in

ExamTech Solutions

localhost:3000/admin

ONLINE EXAM SYSTEM

Sign In

Email

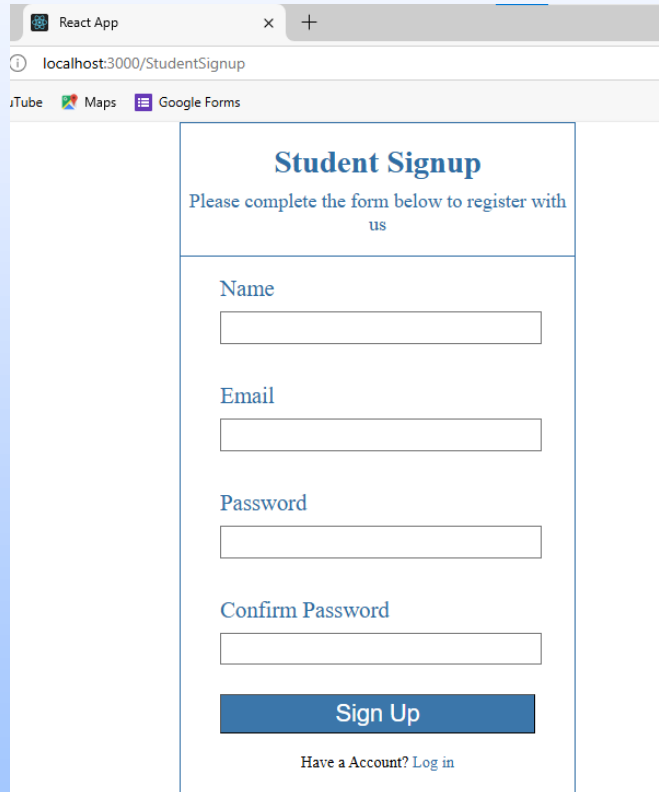
Enter Email

Pasword

Enter Password

Sign in

REGISTER PAGE



React App x +

localhost:3000/StudentSignup

YouTube Maps Google Forms

Student Signup

Please complete the form below to register with us

Name

Email

Password

Confirm Password

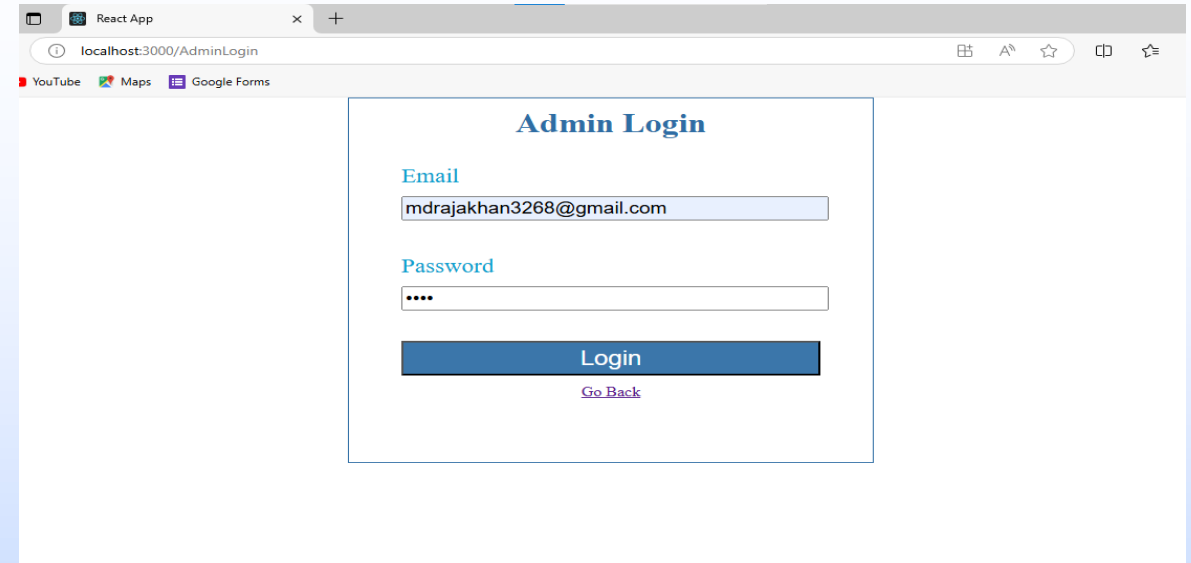
Sign Up

Have a Account? [Log in](#)

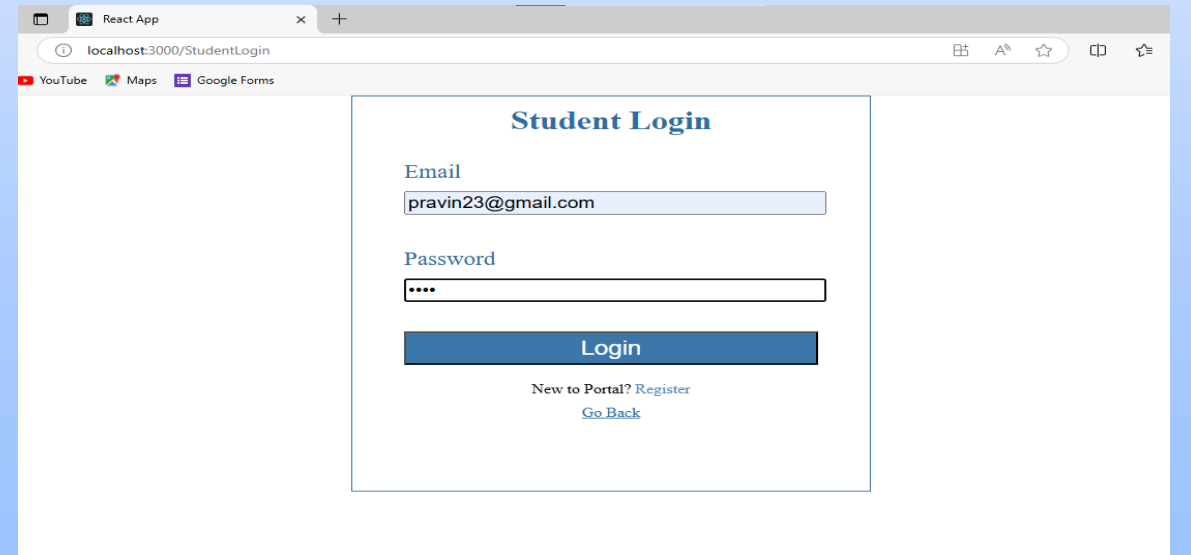
- This Page allow New user to create Account in **Online Examination System**
- To create an Account user need to provide his/her Email ID Followed by Password , For confirming password user need to re-enter his/her Password in “password confirm” Text Field. then by clicking sign up Button user will get his/her Personal Account
- If user has an account, user need to click in login button, which will re-direct the user to login Page

LOGIN PAGE

- This Primary function of this page is to verify a user's Identity and grant them access to their account. this page serve as a gatekeeper, ensuring only authorized user can access the application
- To verify a user's Identity, user need to provide his/her email id followed by Password, then by clicking Login Button user will get the access of application
- If user doesn't have an account, user can get one by clicking on Sign Up button which will re-direct the user to Register Page

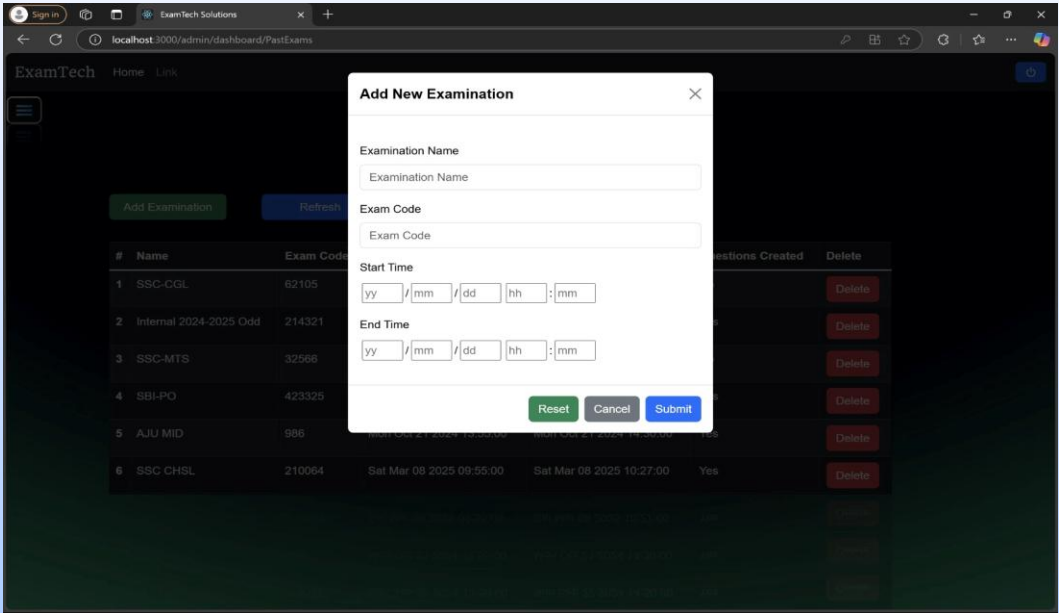
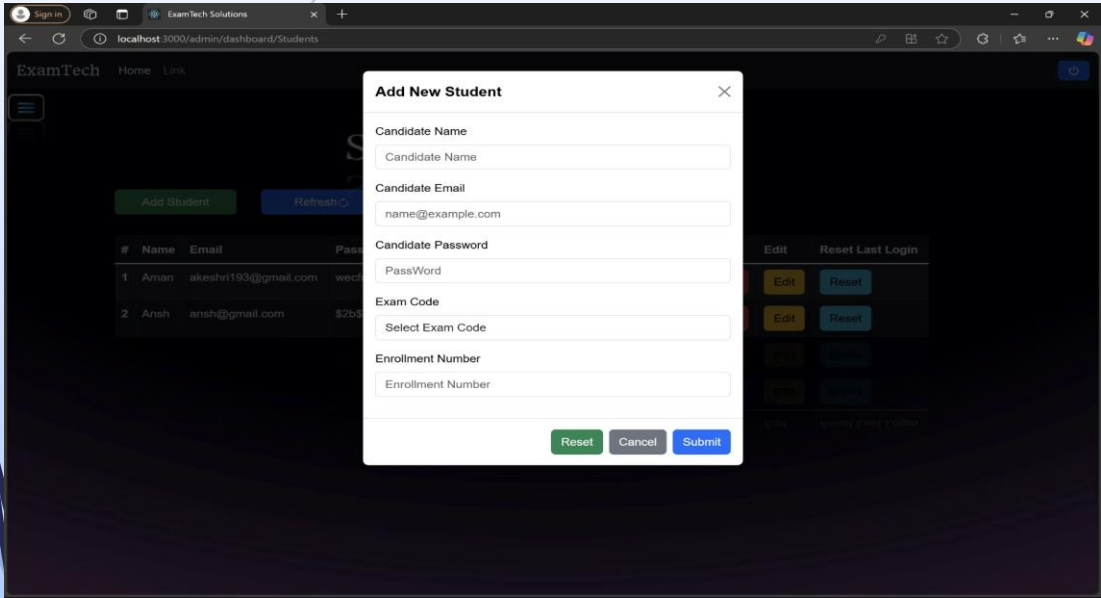


The screenshot shows a web browser window with the title "React App" and the address bar displaying "localhost:3000/AdminLogin". The page content is titled "Admin Login" in a bold, dark blue font. Below the title, there are two input fields: "Email" with the value "mdrajakhan3268@gmail.com" and "Password" with masked characters "....". A dark blue "Login" button is positioned below the password field. At the bottom of the form, there is a link labeled "Go Back" in a smaller, purple font.



The screenshot shows a web browser window with the title "React App" and the address bar displaying "localhost:3000/StudentLogin". The page content is titled "Student Login" in a bold, dark blue font. Below the title, there are two input fields: "Email" with the value "pravin23@gmail.com" and "Password" with masked characters "....". A dark blue "Login" button is positioned below the password field. At the bottom of the form, there is a link labeled "Go Back" in a smaller, purple font, and above it, a link labeled "New to Portal? Register" in a smaller, dark blue font.

SCREENSHOT



SCREENSHOT

The screenshot shows a web application interface titled "CONTROL PANEL" with a mirrored title below it. The interface is dark-themed. At the top left, there is a browser address bar showing "localhost:3000/admin/dashboard/Controlcenter". Below the title, there is an "Exam Code" label above a text input field containing "214321". A blue "Get Details" button is positioned below the input field. A horizontal line separates this section from the next. Below the line is a grey "Hide" button. Underneath the "Hide" button, the text "ExamCode: 214321" is displayed. Below this, there are three buttons: a green "Allow All" button, a white "Allow All" button, and a red "Deny All" button. At the bottom, there are two boxes with a gradient background. The left box is titled "Starting Time" and contains the text "Thu Feb 06 2025 15:30:00 GMT+0530 (India Standard Time)". The right box is titled "Ending Time" and contains the text "Thu Feb 06 2025 16:30:00 GMT+0530 (India Standard Time)".

localhost:3000/admin/dashboard/Controlcenter

CONTROL PANEL

CONTROL PANEL

Exam Code

214321

Get Details

Hide

ExamCode: 214321

Allow All Allow All Deny All

Starting Time

Thu Feb 06 2025 15:30:00
GMT+0530 (India Standard
Time)

Ending Time

Thu Feb 06 2025 16:30:00
GMT+0530 (India Standard
Time)

DATABASE INTEGRATION

MongoDB Compass - localhost:27017/AmanTech

Connections Edit View Help

Compass

My Queries

CONNECTIONS (4)

Search connections

localhost:27017

AmanTech

admins

Storage size:	Documents:	Avg. document size:	Indexes:	Total index size:
20.48 kB	1	147.00 B	1	20.48 kB

disqualifieds

Storage size:	Documents:	Avg. document size:	Indexes:	Total index size:
8.19 kB	0	0 B	2	49.15 kB

examinations

Storage size:	Documents:	Avg. document size:	Indexes:	Total index size:
20.48 kB	5	122.00 B	3	110.59 kB

publishes

Storage size:	Documents:	Avg. document size:	Indexes:	Total index size:
20.48 kB	2	61.00 B	1	36.86 kB

questions

Storage size:	Documents:	Avg. document size:	Indexes:	Total index size:
20.48 kB	2	185.00 B	2	73.73 kB

MongoDB Compass - localhost:27017/AmanTech.examinations

Connections Edit View Collection Help

Compass

My Queries

CONNECTIONS (4)

Search connections

localhost:27017

AmanTech

examinations

Documents 5 Aggregations Schema Indexes 3 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE

25 1-5 of 5

```
{ "_id": ObjectId("66ee862fb77985bd4e55f127"),  
  "Name": "SSC-CGL",  
  "examcode": 62105,  
  "start": 1735899300000,  
  "end": 1735812900000,  
  "__v": 0,  
  "quesCreated": "No"  
}
```

```
{ "_id": ObjectId("66ee8880eba447b534843225"),  
  "Name": "Internal 2024-2025 Odd",  
  "examcode": 214321,  
  "start": 1738836000000,  
  "end": 1738839600000,  
  "__v": 0,  
  "quesCreated": "Yes",  
  "SessCreated": "Yes"  
}
```

```
{ "_id": ObjectId("66efc7887a203e68a25e5093"),  
  "Name": "SSC-MTS",  
  "examcode": 32566,  
  "start": 1712052600000,  
  "end": 1712056200000,  
  "__v": 0,  
  "quesCreated": "No"  
}
```

```
{ "_id": ObjectId("66efd50a6bae19448d836df8"),  
  "Name": "SBI-PO"  
}
```

NEW FEATURES

MULTIPLE TAB DETECTION

SPLIT SCREEN DETECTION

DEVTOOLS BLOCKER

DISABLED RIGHT MOUSE CLICK

LIVE SYSTEM FEED

SOCKET SERVER-LIVE DATA UPDATE SECOND BY SECOND

AUTOMATIC LOGOUT IF CANDIDATE DETECTED CHEATING

DESIGN ENHANCEMENT AND UPDATE

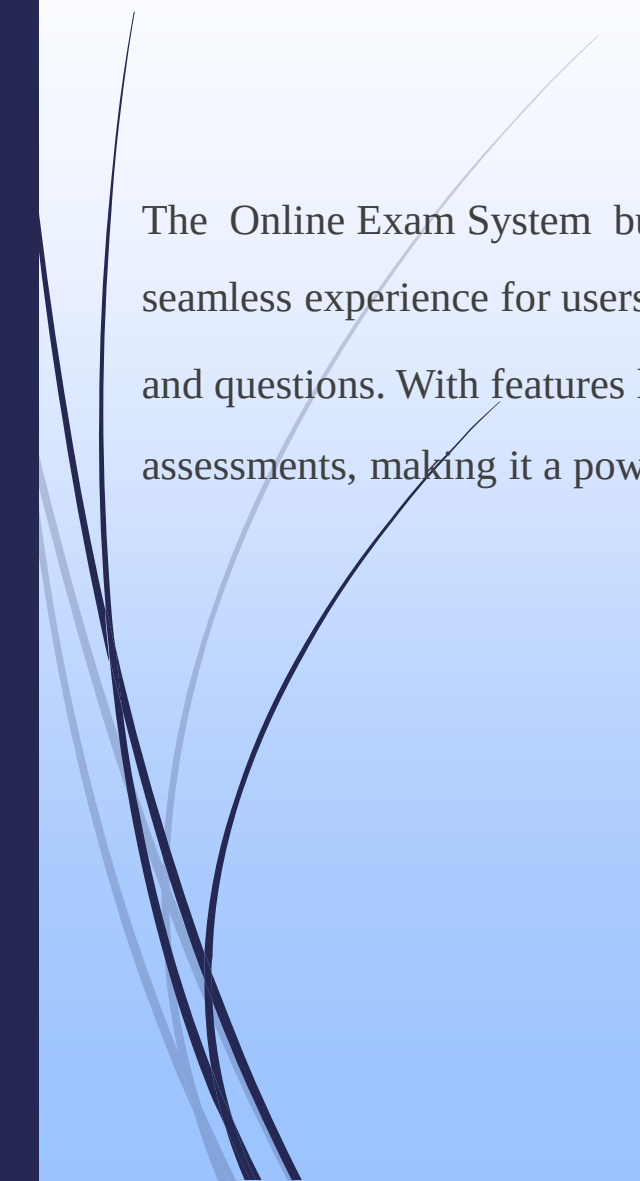
- ❑ **MULTIPLE TAB DETECTION** : Multiple tab detection have the ability of the system to detect and prevent students from opening multiple tabs or windows during an exam.
- ❑ **SPLIT SCREEN DETECTION** : Split screen detection is used when a user is attempting to divide their screen into multiple sections or windows, potentially to access unauthorized resources or cheat.
- ❑ **DEVTOOLS BLOCKER** : DevTools Blocker is a feature or tool used in online exam systems and other secure environments to prevent users from accessing the browser's developer tools during a session.
- ❑ **DISABLED RIGHT MOUSE CLICK** : Disabled the right mouse click is a feature used in online exam systems and other secure environments to prevent users from accessing certain functionalities or information.
- ❑ **LIVE SYSTEM FEED** : A live system feed refers to the real-time monitoring and display of system-related information, such as Real-time display of system logs, including error messages, warnings, and other system events.



- ❑ **SOCKET SERVER-LIVE DATA UPDATE SECOND BY SECOND** : It Establish a persistent, low-latency connection between the client and server , providing features like automatic reconnection and multiplexing.
- ❑ **AUTOMATIC LOGOUT IF CANDITATE DETECTED CHEATING** :Automatic logout if a candidate is detected cheating is a mechanism used in online exam systems to prevent and detect cheating attempts
- ❑ **DESIGN ENHANCEMENT AND UPDATE** : Design enhancement and update refers to the process of refining and improving the visual and functional aspects of a product, system, or application.



CONCLUSION



The Online Exam System built using the MERN stack efficiently streamlines exam management and participation. It offers a seamless experience for users to take tests and view results, while admins have full control over managing subjects, exams, and questions. With features like result tracking and exam oversight, this system enhances the overall process of online assessments, making it a powerful tool for both educators and students.